RA & SQL Practice Questions

200 Questions for Exam Prep
Database Systems - CPH

Symbol Reference

Π	Project	pi
σ	Select	sigma
ρ	Rename	rho
γ	Aggregation	gamma
$\bowtie$	Natural Join	join
$\times$	Cartesian Product	x
$\cup$	Union	union
$\cap$	Intersection	intersect
$-$	Difference	minus

Schema Reference

Hospital: patient(patientID, name, age, gender, bloodType), doctor(docID, name, specialty, salary), appointment(appointID, patientID, docID, appointDate), prescription(prescID, appointID, medicine, dosage)

Library: book(bookID, title, author, genre, publishYear), member(memberID, name, email, joinDate), loan(loanID, bookID, memberID, loanDate, returnDate), fine(fineID, loanID, amount, paidStatus)

E-Commerce: customer(custID, name, email, city, country), product(prodID, name, category, price, stock), orders(orderID, custID, orderDate, totalAmount), orderItem(orderID, prodID, quantity, unitPrice)

Part 1: Relational Algebra (100 Questions)

1 Basic Selection and Projection

Q1. Find all patients older than 60. List patientID and name.

Answer

$\Pi_{\text{patientID, name}}(\sigma_{\text{age}>60}(\text{patient}))$

Q2. Find all doctors with specialty "Cardiology". List docID and name.

Answer

$\Pi_{\text{docID, name}}(\sigma_{\text{specialty}=\text{"Cardiology"}}(\text{doctor}))$

Q3. Find all female patients with blood type "O+". List name and age.

Answer

$\Pi_{\text{name, age}}(\sigma_{\text{gender}=\text{"F"}\wedge\text{bloodType}=\text{"O+"}}(\text{patient}))$

Q4. Find all doctors earning more than 100000. List only names.

Answer

$\Pi_{\text{name}}(\sigma_{\text{salary}>100000}(\text{doctor}))$

Q5. Find all unique specialties in the doctor table.

Answer

$\Pi_{\text{specialty}}(\text{doctor})$

Q6. Find patients aged between 30 and 50 (inclusive). List patientID.

Answer

$\Pi_{\text{patientID}}(\sigma_{\text{age}\geq 30\wedge\text{age}\leq 50}(\text{patient}))$

Q7. Find all prescriptions with dosage "500mg". List prescID and medicine.

Answer

$\Pi_{\text{prescID, medicine}}(\sigma_{\text{dosage}=\text{"500mg"}}(\text{prescription}))$

Q8. Find all appointments on "2024-05-15". List appointID.

Answer

$\Pi_{\text{appointID}}(\sigma_{\text{appointDate}=\text{"2024-05-15"}}(\text{appointment}))$

Q9. Find patients who are NOT male. List patientID and name.

Answer

$\Pi_{\text{patientID, name}}(\sigma_{\text{gender}\neq\text{"M"}}(\text{patient}))$

Q10. Find doctors who are "Surgeon" OR "Neurologist". List docID.

Answer

$\Pi_{\text{docID}}(\sigma_{\text{specialty}=\text{“Surgeon”}\vee\text{specialty}=\text{“Neurologist”}}(\text{doctor}))$

Q11. Find all books published after 2020. List bookID and title.

Answer

$\Pi_{\text{bookID, title}}(\sigma_{\text{publishYear}>2020}(\text{book}))$

Q12. Find all books in “Science Fiction” genre. List title.

Answer

$\Pi_{\text{title}}(\sigma_{\text{genre}=\text{“Science Fiction”}}(\text{book}))$

Q13. Find all unpaid fines. List fineID and amount.

Answer

$\Pi_{\text{fineID, amount}}(\sigma_{\text{paidStatus}=\text{“unpaid”}}(\text{fine}))$

Q14. Find all unique genres in the book table.

Answer

$\Pi_{\text{genre}}(\text{book})$

Q15. Find all customers from “Denmark”. List custID and name.

Answer

$\Pi_{\text{custID, name}}(\sigma_{\text{country}=\text{“Denmark”}}(\text{customer}))$

Q16. Find all products with price over 500. List prodID and name.

Answer

$\Pi_{\text{prodID, name}}(\sigma_{\text{price}>500}(\text{product}))$

Q17. Find products in “Electronics” with stock less than 10.

Answer

$\Pi_{\text{prodID, name}}(\sigma_{\text{category}=\text{“Electronics”}\wedge\text{stock}<10}(\text{product}))$

Q18. Find all orders with totalAmount exceeding 1000. List orderID.

Answer

$\Pi_{\text{orderID}}(\sigma_{\text{totalAmount}>1000}(\text{orders}))$

Q19. Find all unique categories in the product table.

Answer

$\Pi_{\text{category}}(\text{product})$

Q20. Find products priced between 100 and 500. List name.

Answer

$\Pi_{\text{name}}(\sigma_{\text{price}\geq 100\wedge\text{price}\leq 500}(\text{product}))$

2 Join Operations

Q21. Find names of patients who have at least one appointment.

Answer

$\Pi_{\text{patient.name}}(\text{patient} \bowtie \text{appointment})$

Q22. Find patient names treated by a Cardiologist.

Answer

$\Pi_{\text{patient.name}}(\text{patient} \bowtie \text{appointment} \bowtie (\sigma_{\text{specialty}=\text{“Cardiology”}}(\text{doctor})))$

Q23. Find medicines prescribed to patients older than 70.

Answer

$\Pi_{\text{medicine}}(\text{prescription} \bowtie \text{appointment} \bowtie (\sigma_{\text{age}>70}(\text{patient})))$

Q24. Find doctor names with appointments on “2024-06-01”.

Answer

$\Pi_{\text{doctor.name}}(\text{doctor} \bowtie (\sigma_{\text{appointDate}=\text{“2024-06-01”}}(\text{appointment})))$

Q25. Find patient names and their prescribed medicines.

Answer

$\Pi_{\text{patient.name, medicine}}(\text{patient} \bowtie \text{appointment} \bowtie \text{prescription})$

Q26. Find appointID where doctor salary exceeds 80000.

Answer

$\Pi_{\text{appointID}}(\text{appointment} \bowtie (\sigma_{\text{salary}>80000}(\text{doctor})))$

Q27. Find prescriptions for “Aspirin” with patient names.

Answer

$\Pi_{\text{patient.name, prescID}}((\sigma_{\text{medicine}=\text{“Aspirin”}}(\text{prescription})) \bowtie \text{appointment} \bowtie \text{patient})$

Q28. Find female patients who saw a surgeon. List names.

Answer

$\Pi_{\text{patient.name}}((\sigma_{\text{gender}=\text{“F”}}(\text{patient})) \bowtie \text{appointment} \bowtie (\sigma_{\text{specialty}=\text{“Surgeon”}}(\text{doctor})))$

Q29. Find doctor and patient name pairs for appointments with prescriptions.

Answer

$\Pi_{\text{doctor.name, patient.name}}(\text{doctor} \bowtie \text{appointment} \bowtie \text{prescription} \bowtie \text{patient})$

Q30. Find patientID treated by doctors earning over 90000.

Answer

$\Pi_{\text{patientID}}(\text{appointment} \bowtie (\sigma_{\text{salary}>90000}(\text{doctor})))$

Q31. Find member names who borrowed at least one book.

Answer

$\Pi_{\text{member.name}}(\text{member} \bowtie \text{loan})$

Q32. Find book titles borrowed by member "Alice".

Answer

$\Pi_{\text{book.title}}(\text{book} \bowtie \text{loan} \bowtie (\sigma_{\text{name="Alice"}}(\text{member})))$

Q33. Find member names with unpaid fines.

Answer

$\Pi_{\text{member.name}}(\text{member} \bowtie \text{loan} \bowtie (\sigma_{\text{paidStatus="unpaid"}}(\text{fine})))$

Q34. Find book titles and member names for all loans.

Answer

$\Pi_{\text{book.title, member.name}}(\text{book} \bowtie \text{loan} \bowtie \text{member})$

Q35. Find members with fines over 50. List memberID.

Answer

$\Pi_{\text{memberID}}(\text{loan} \bowtie (\sigma_{\text{amount}>50}(\text{fine})))$

Q36. Find customer names who placed at least one order.

Answer

$\Pi_{\text{customer.name}}(\text{customer} \bowtie \text{orders})$

Q37. Find product names ordered by customers from "Copenhagen".

Answer

$\Pi_{\text{product.name}}(\text{product} \bowtie \text{orderItem} \bowtie \text{orders} \bowtie (\sigma_{\text{city="Copenhagen"}}(\text{customer})))$

Q38. Find orderID containing "Electronics" products.

Answer

$\Pi_{\text{orderID}}(\text{orderItem} \bowtie (\sigma_{\text{category="Electronics"}}(\text{product})))$

Q39. Find customer and product name pairs for all orders.

Answer

$\Pi_{\text{customer.name, product.name}}(\text{customer} \bowtie \text{orders} \bowtie \text{orderItem} \bowtie \text{product})$

Q40. Find prodID ordered with quantity greater than 5.

Answer

$\Pi_{\text{prodID}}(\sigma_{\text{quantity}>5}(\text{orderItem}))$

3 Set Difference - "NOT" Queries

Q41. Find patients who have NOT had any appointment. List patientID.

Answer

$\Pi_{\text{patientID}}(\text{patient}) - \Pi_{\text{patientID}}(\text{appointment})$

Q42. Find doctors who have NOT seen any patients. List docID.

Answer

$\Pi_{\text{docID}}(\text{doctor}) - \Pi_{\text{docID}}(\text{appointment})$

Q43. Find patients never prescribed "Ibuprofen". List patientID.

Answer

$\Pi_{\text{patientID}}(\text{patient}) - \Pi_{\text{patientID}}(\text{appointment} \bowtie (\sigma_{\text{medicine}=\text{"Ibuprofen"}}(\text{prescription})))$

Q44. Find doctors who have NOT seen female patients. List docID.

Answer

$\Pi_{\text{docID}}(\text{doctor}) - \Pi_{\text{docID}}(\text{appointment} \bowtie (\sigma_{\text{gender}=\text{"F"}}(\text{patient})))$

Q45. Find appointments without prescriptions. List appointID.

Answer

$\Pi_{\text{appointID}}(\text{appointment}) - \Pi_{\text{appointID}}(\text{prescription})$

Q46. Find books never borrowed. List bookID.

Answer

$\Pi_{\text{bookID}}(\text{book}) - \Pi_{\text{bookID}}(\text{loan})$

Q47. Find members who never borrowed "Mystery" books. List memberID.

Answer

$\Pi_{\text{memberID}}(\text{member}) - \Pi_{\text{memberID}}(\text{loan} \bowtie (\sigma_{\text{genre}=\text{"Mystery"}}(\text{book})))$

Q48. Find loans without fines. List loanID.

Answer

$\Pi_{\text{loanID}}(\text{loan}) - \Pi_{\text{loanID}}(\text{fine})$

Q49. Find members who never had a fine. List memberID.

Answer

$\Pi_{\text{memberID}}(\text{member}) - \Pi_{\text{memberID}}(\text{loan} \bowtie \text{fine})$

Q50. Find authors whose books have never been borrowed.

Answer

$\Pi_{\text{author}}(\text{book}) - \Pi_{\text{author}}(\text{book} \bowtie \text{loan})$

Q51. Find customers who have NOT placed any orders. List custID.

Answer

$\Pi_{\text{custID}}(\text{customer}) - \Pi_{\text{custID}}(\text{orders})$

Q52. Find products never ordered. List prodID.

Answer

$\Pi_{\text{prodID}}(\text{product}) - \Pi_{\text{prodID}}(\text{orderItem})$

Q53. Find customers who have NOT ordered "Clothing". List custID.

Answer

$\Pi_{\text{custID}}(\text{customer}) - \Pi_{\text{custID}}(\text{orders} \bowtie \text{orderItem} \bowtie (\sigma_{\text{category}=\text{"Clothing"}}(\text{product})))$

Q54. Find patients with appointments but no prescriptions. List patientID.

Answer

$\Pi_{\text{patientID}}(\text{appointment}) - \Pi_{\text{patientID}}(\text{appointment} \bowtie \text{prescription})$

Q55. Find doctors who treated only male patients. List docID.

Answer

$\Pi_{\text{docID}}(\text{appointment}) - \Pi_{\text{docID}}(\text{appointment} \bowtie (\sigma_{\text{gender}\neq\text{"M"}}(\text{patient})))$

4 Set Operations - Union and Intersection

Q56. Find patients who saw BOTH Cardiologist AND Surgeon. List patientID.

Answer

$\Pi_{\text{patientID}}(\text{appointment} \bowtie (\sigma_{\text{specialty}=\text{"Cardiology"}}(\text{doctor}))) \cap \Pi_{\text{patientID}}(\text{appointment} \bowtie (\sigma_{\text{specialty}=\text{"Surgeon"}}(\text{doctor})))$

Q57. Find patients who saw Cardiologist OR Surgeon. List patientID.

Answer

$\Pi_{\text{patientID}}(\text{appointment} \bowtie (\sigma_{\text{specialty}=\text{"Cardiology"}}(\text{doctor}))) \cup \Pi_{\text{patientID}}(\text{appointment} \bowtie (\sigma_{\text{specialty}=\text{"Surgeon"}}(\text{doctor})))$

Q58. Find members who borrowed BOTH "Mystery" AND "Romance". List memberID.

Answer

$\Pi_{\text{memberID}}(\text{loan} \bowtie (\sigma_{\text{genre}=\text{"Mystery"}}(\text{book}))) \cap \Pi_{\text{memberID}}(\text{loan} \bowtie (\sigma_{\text{genre}=\text{"Romance"}}(\text{book})))$

Q59. Find customers who ordered BOTH "Electronics" AND "Clothing". List custID.

Answer

$\Pi_{\text{custID}}(\text{orders} \bowtie \text{orderItem} \bowtie (\sigma_{\text{category}=\text{"Electronics"}}(\text{product}))) \cap \Pi_{\text{custID}}(\text{orders} \bowtie \text{orderItem} \bowtie (\sigma_{\text{category}=\text{"Clothing"}}(\text{product})))$

Q60. Find doctors who treated "A+" OR "B+" patients. List docID.

Answer

$\Pi_{\text{docID}}(\text{appointment} \bowtie (\sigma_{\text{bloodType}=\text{"A+"}}(\text{patient}))) \cup \Pi_{\text{docID}}(\text{appointment} \bowtie (\sigma_{\text{bloodType}=\text{"B+"}}(\text{patient})))$

5 Aggregation Queries

Q61. Count total number of patients.

Answer

$\gamma_{;\text{count}(*)}(\text{patient})$

Q62. Find average salary of doctors by specialty.

Answer

$\gamma_{\text{specialty};\text{avg}(\text{salary})}(\text{doctor})$

Q63. Find maximum fine amount.

Answer

$\gamma_{;\text{max}(\text{amount})}(\text{fine})$

Q64. Count books per genre.

Answer

$\gamma_{\text{genre};\text{count}(*)}(\text{book})$

Q65. Find min and max price per category.

Answer

$\gamma_{\text{category};\text{min}(\text{price}),\ \text{max}(\text{price})}(\text{product})$

Q66. Find total quantity ordered per product.

Answer

$\gamma_{\text{prodID};\text{sum}(\text{quantity})}(\text{orderItem})$

Q67. Find average age by gender.

Answer

$\gamma_{\text{gender};\text{avg}(\text{age})}(\text{patient})$

Q68. Count orders per customer.

Answer

$\gamma_{\text{custID};\text{count}(*)}(\text{orders})$

Q69. Find sum of unpaid fines.

Answer

$\gamma_{;\text{sum}(\text{amount})}(\sigma_{\text{paidStatus}=\text{“unpaid”}}(\text{fine}))$

Q70. Count appointments per doctor.

Answer

$\gamma_{\text{docID};\text{count}(*)}(\text{appointment})$

6 Rename and Self-Join

Q71. Find pairs of patients with same blood type.

Answer

$\Pi_{\text{p1.name, p2.name}}(\sigma_{\text{p1.bloodType=p2.bloodType} \wedge \text{p1.patientID<p2.patientID}}(\rho_{\text{p1}}(\text{patient}) \times \rho_{\text{p2}}(\text{patient})))$

Q72. Find pairs of doctors with same specialty.

Answer

$\Pi_{\text{d1.name, d2.name}}(\sigma_{\text{d1.specialty=d2.specialty} \wedge \text{d1.docID<d2.docID}}(\rho_{\text{d1}}(\text{doctor}) \times \rho_{\text{d2}}(\text{doctor})))$

Q73. Find pairs of books by same author.

Answer

$\Pi_{\text{b1.title, b2.title}}(\sigma_{\text{b1.author=b2.author} \wedge \text{b1.bookID<b2.bookID}}(\rho_{\text{b1}}(\text{book}) \times \rho_{\text{b2}}(\text{book})))$

Q74. Find customers in same city.

Answer

$\Pi_{\text{c1.name, c2.name}}(\sigma_{\text{c1.city=c2.city} \wedge \text{c1.custID<c2.custID}}(\rho_{\text{c1}}(\text{customer}) \times \rho_{\text{c2}}(\text{customer})))$

Q75. Find doctors earning more than another in same specialty.

Answer

$\Pi_{\text{d1.name}}(\sigma_{\text{d1.salary>d2.salary} \wedge \text{d1.specialty=d2.specialty}}(\rho_{\text{d1}}(\text{doctor}) \times \rho_{\text{d2}}(\text{doctor})))$

7 Complex Queries

Q76. Find patients with appointments with ALL cardiologists. (Division)

Answer

$\Pi_{\text{patientID, docID}}(\text{appointment} \bowtie (\sigma_{\text{specialty="Cardiology"}}(\text{doctor}))) \div \Pi_{\text{docID}}(\sigma_{\text{specialty="Cardiology"}}(\text{doctor}))$

Q77. Find members who borrowed ALL genres.

Answer

$\Pi_{\text{memberID, genre}}(\text{loan} \bowtie \text{book}) \div \Pi_{\text{genre}}(\text{book})$

Q78. Find specialties where ALL doctors earn over 70000.

Answer

$\Pi_{\text{specialty}}(\text{doctor}) - \Pi_{\text{specialty}}(\sigma_{\text{salary} \leq 70000}(\text{doctor}))$

Q79. Find genres where ALL books published after 2015.

Answer

$\Pi_{\text{genre}}(\text{book}) - \Pi_{\text{genre}}(\sigma_{\text{publishYear} \leq 2015}(\text{book}))$

Q80. Find patients who saw exactly one doctor.

Answer

$\Pi_{\text{patientID}}(\text{appointment}) - \Pi_{\text{a1.patientID}}(\sigma_{\text{a1.docID}\neq\text{a2.docID}}(\rho_{\text{a1}}(\text{appointment}) \bowtie_{\text{a1.patientID}=\text{a2.patientID}} \rho_{\text{a2}}(\text{appointment})))$

Q81. Find medicines for patients with blood type "AB-".

Answer

$\Pi_{\text{medicine}}(\text{prescription} \bowtie \text{appointment} \bowtie (\sigma_{\text{bloodType}=\text{"AB-"}}(\text{patient})))$

Q82. Find customers from "Germany" who ordered products over 100.

Answer

$\Pi_{\text{customer.name}}((\sigma_{\text{country}=\text{"Germany"}}(\text{customer})) \bowtie \text{orders} \bowtie \text{orderItem} \bowtie (\sigma_{\text{price}>100}(\text{product})))$

Q83. Find patients prescribed "Ibuprofen" but NOT "Aspirin".

Answer

$\Pi_{\text{patientID}}(\text{appointment} \bowtie (\sigma_{\text{medicine}=\text{"Ibuprofen"}}(\text{prescription}))) - \Pi_{\text{patientID}}(\text{appointment} \bowtie (\sigma_{\text{medicine}=\text{"Aspirin"}}(\text{prescription})))$

Q84. Find books borrowed by members without fines.

Answer

$\Pi_{\text{bookID}}(\text{loan} \bowtie (\Pi_{\text{memberID}}(\text{member}) - \Pi_{\text{memberID}}(\text{loan} \bowtie \text{fine})))$

Q85. Find members who borrowed at least 2 different books.

Answer

$\Pi_{\text{memberID}}(\sigma_{\text{l1.bookID}\neq\text{l2.bookID}}(\rho_{\text{l1}}(\text{loan}) \bowtie_{\text{l1.memberID}=\text{l2.memberID}} \rho_{\text{l2}}(\text{loan})))$

Q86. Find customers who ordered from at least 2 categories.

Answer

$\Pi_{\text{custID}}(\sigma_{\text{p1.category}\neq\text{p2.category}}((\text{orders} \bowtie \text{orderItem} \bowtie \rho_{\text{p1}}(\text{product})) \bowtie (\text{orders} \bowtie \text{orderItem} \bowtie \rho_{\text{p2}}(\text{product}))))$

Q87. Find members who borrowed ALL J.K. Rowling books.

Answer

$\Pi_{\text{memberID, bookID}}(\text{loan} \bowtie (\sigma_{\text{author}=\text{"J.K. Rowling"}}(\text{book}))) \div \Pi_{\text{bookID}}(\sigma_{\text{author}=\text{"J.K. Rowling"}}(\text{book}))$

Q88. Find products ordered by customers from ALL countries.

Answer

$\Pi_{\text{prodID, country}}(\text{orderItem} \bowtie \text{orders} \bowtie \text{customer}) \div \Pi_{\text{country}}(\text{customer})$

Q89. Find doctors who treated patients younger than 18.

Answer

$\Pi_{\text{docID}}(\text{appointment} \bowtie (\sigma_{\text{age}<18}(\text{patient})))$

Q90. Find orders containing ONLY "Electronics" products.

Answer

$\Pi_{\text{orderID}}(\text{orderItem} \bowtie (\sigma_{\text{category}=\text{“Electronics”}}(\text{product}))) - \Pi_{\text{orderID}}(\text{orderItem} \bowtie (\sigma_{\text{category}\neq\text{“Electronics”}}(\text{product})))$

Q91. Find customers who ordered same product multiple times.

Answer

$\Pi_{\text{custID}}(\sigma_{\text{oi1.orderID}\neq\text{oi2.orderID}}(\rho_{\text{oi1}}(\text{orderItem} \bowtie \text{orders}) \bowtie_{\text{oi1.prodID}=\text{oi2.prodID}\wedge\text{oi1.custID}=\text{oi2.custID}} \rho_{\text{oi2}}(\text{orderItem} \bowtie \text{orders})))$

Q92. Find patients treated by all surgeons.

Answer

$\Pi_{\text{patientID, docID}}(\text{appointment} \bowtie (\sigma_{\text{specialty}=\text{“Surgeon”}}(\text{doctor}))) \div \Pi_{\text{docID}}(\sigma_{\text{specialty}=\text{“Surgeon”}}(\text{doctor}))$

Q93. Find customers who ordered ALL “Electronics” products.

Answer

$\Pi_{\text{custID, prodID}}(\text{orders} \bowtie \text{orderItem} \bowtie (\sigma_{\text{category}=\text{“Electronics”}}(\text{product}))) \div \Pi_{\text{prodID}}(\sigma_{\text{category}=\text{“Electronics”}}(\text{product}))$

Q94. Find categories with products out of stock (stock=0).

Answer

$\Pi_{\text{category}}(\sigma_{\text{stock}=0}(\text{product}))$

Q95. Find doctors who treated all blood types.

Answer

$\Pi_{\text{docID, bloodType}}(\text{appointment} \bowtie \text{patient}) \div \Pi_{\text{bloodType}}(\text{patient})$

Q96. Find members who borrowed books from “Mystery” genre lovers.

Answer

$\Pi_{\text{bookID}}(\text{loan} \bowtie (\Pi_{\text{memberID}}(\text{loan} \bowtie (\sigma_{\text{genre}=\text{“Mystery”}}(\text{book})))))$

Q97. Find patients on same date with different doctors.

Answer

$\Pi_{\text{patientID}}(\sigma_{\text{a1.docID}\neq\text{a2.docID}\wedge\text{a1.appointDate}=\text{a2.appointDate}}(\rho_{\text{a1}}(\text{appointment}) \bowtie_{\text{a1.patientID}=\text{a2.patientID}} \rho_{\text{a2}}(\text{appointment})))$

Q98. Find books borrowed more than once by same member.

Answer

$\Pi_{\text{bookID}}(\sigma_{\text{l1.loanID}\neq\text{l2.loanID}}(\rho_{\text{l1}}(\text{loan}) \bowtie_{\text{l1.bookID}=\text{l2.bookID}\wedge\text{l1.memberID}=\text{l2.memberID}} \rho_{\text{l2}}(\text{loan})))$

Q99. Find unique countries with customers.

Answer

$\Pi_{\text{country}}(\text{customer})$

Q100. Find products never ordered by Danish customers.

Answer

$\Pi_{\text{prodID}}(\text{product}) - \Pi_{\text{prodID}}(\text{orderItem} \bowtie \text{orders} \bowtie (\sigma_{\text{country}=\text{“Denmark”}}(\text{customer})))$

Part 2: SQL (100 Questions)

8 True/False (Q1-Q20)

Q1. BETWEEN is inclusive of both endpoints.

Answer

True — BETWEEN 10 AND 20 includes 10 and 20.

Q2. COUNT(column) includes NULL values.

Answer

False — COUNT(col) excludes NULL. COUNT(*) includes all rows.

Q3. NATURAL JOIN keeps one copy of common columns.

Answer

True

Q4. ORDER BY executes before GROUP BY.

Answer

False — FROM → WHERE → GROUP BY → HAVING → SELECT → ORDER BY

Q5. Subquery with IN can return multiple rows.

Answer

True

Q6. UNION ALL removes duplicates.

Answer

False — UNION removes dups; UNION ALL keeps all.

Q7. FK can reference UNIQUE column (not just PK).

Answer

True

Q8. 5 * NULL = 0.

Answer

False — Any arithmetic with NULL = NULL.

Q9. LEFT JOIN returns all left rows even without match.

Answer

True

Q10. GROUP BY must appear before HAVING.

Answer

True

Q11. Views with aggregates are updatable.

Answer

False

Q12. x > SOME means x > at least one value.

Answer

True

Q13. COALESCE returns first non-NULL.

Answer

True

Q14. ALTER TABLE is DML.

Answer

False — It's DDL.

Q15. x IN (empty) = FALSE.

Answer

True

Q16. Correlated subquery references outer query.

Answer

True

Q17. TRUNCATE removes rows but keeps structure.

Answer

True

Q18. Aggregates allowed in WHERE.

Answer

False — Use HAVING.

Q19. NOT EXISTS = TRUE when subquery empty.

Answer

True

Q20. DISTINCT applies to all SELECT columns.

Answer

True

9 Multiple Choice (Q21-Q60)

Q21. Find patients older than 50 with blood type 'O+'.
(a) `WHERE age > 50 AND bloodType = 'O+'` (b) `WHERE age > 50 OR bloodType = 'O+'`

Answer

(a)

Q22. Handle NULL in WHERE:
(a) `= NULL` (b) `IS NULL`

Answer

(b) — Always use IS NULL.

Q23. Find names starting with 'Dr.':
(a) `name = 'Dr.%'` (b) `name LIKE 'Dr.%'`

Answer

(b)

Q24. Unique specialties: (a) UNIQUE (b) DISTINCT (c) GROUP BY (d) b and c

Answer

(d)

Q25. Patients aged 20, 25, or 30: (a) age=20 AND age=25 (b) age IN (20,25,30)

Answer

(b)

Q26. Order salary DESC, name ASC: (a) ORDER BY salary, name DESC (b) ORDER BY salary DESC, name ASC

Answer

(b)

Q27. SELECT * FROM A, B returns: (a) Natural join (b) Cartesian product

Answer

(b)

Q28. 5 + NULL = ? (a) 5 (b) 0 (c) NULL

Answer

(c)

Q29. Average salary per specialty:
(a) `SELECT specialty, AVG(salary) FROM doctor`
(b) `SELECT specialty, AVG(salary) FROM doctor GROUP BY specialty`

Answer

(b)

Q30. Specialties with AVG(salary) > 80000: (a) WHERE (b) HAVING

Answer

(b) — HAVING for aggregate conditions.

Q31. AVG of (25, NULL, 30, NULL, 25) = ? (a) 16 (b) 26.67

Answer

(b) — AVG ignores NULL: (25+30+25)/3.

Q32. Count non-NULL bloodType: (a) COUNT(*) (b) COUNT(bloodType)

Answer

(b)

Q33. SUM when no rows match: (a) 0 (b) NULL

Answer

(b)

Q34. Second highest salary approach: subquery with MAX < (SELECT MAX) or ORDER BY LIMIT OFFSET.

Answer

Both work.

Q35. Equivalent to A, B WHERE A.id=B.id: (a) NATURAL JOIN (b) JOIN ON

Answer

(b) — JOIN ON is equivalent; NATURAL may join other columns.

Q36. RIGHT JOIN preserves: (a) left (b) right

Answer

(b)

Q37. 4 rows CROSS JOIN 3 rows = ? (a) 7 (b) 12

Answer

(b) — $4 \times 3 = 12$.

Q38. Find patients with NO appointments: LEFT JOIN + IS NULL or NOT IN.

Answer

Both work.

Q39. NATURAL JOIN with no common columns: (a) Error (b) Cartesian

Answer

(b)

Q40. JOIN USING vs ON: USING keeps one copy of join column.

Answer

True

Q41. Patients treated by Cardiologist: nested IN subqueries.

Answer

Use IN for multiple rows from subquery.

Q42. x > ALL means: (a) > at least one (b) > every value

Answer

(b)

Q43. x > ALL (empty) = ? (a) FALSE (b) TRUE

Answer

(b) — Vacuously true.

Q44. Patients NOT seen by Surgeon: NOT IN (SELECT patientID FROM appointment WHERE docID IN (surgeons)).

Answer

Correct pattern.

Q45. Check existence: EXISTS, IN, COUNT > 0 all work.

Answer

All valid.

Q46. EXCEPT does: A - B (rows in first not second).

Answer

Correct.

Q47. BOTH A AND B: use INTERSECT.

Answer

Correct.

Q48. Scalar subquery returns: one row, one column.

Answer

Correct.

Q49. Correlated subquery: references outer query values.

Answer

Correct.

Q50. UNION requires: same columns, compatible types.

Answer

Correct.

Q51. Salary non-negative: CHECK(salary >= 0).

Answer

Correct.

Q52. ON UPDATE CASCADE: updates FK to match new PK.

Answer

Correct.

Q53. DROP vs TRUNCATE: DROP removes structure; TRUNCATE keeps it.

Answer

Correct.

Q54. Composite PK: PRIMARY KEY (col1, col2).

Answer

Correct.

Q55. ROLLBACK: undoes all transaction changes.

Answer

Correct.

Q56. Add column: ALTER TABLE ADD COLUMN.

Answer

Correct.

Q57. UNIQUE allows NULL (unlike PK).

Answer

True.

Q58. ON DELETE SET NULL: sets FK to NULL when parent deleted.

Answer

Correct.

Q59. SAVEPOINT: marks point for partial rollback.

Answer

Correct.

Q60. CASE WHEN syntax: CASE WHEN cond THEN val ELSE default END.

Answer

Correct.

10 Fill-in-the-Box (Q61-Q100)

Q61. Doctors earning more than average:

```
WHERE salary > ([Box]);
```

Answer

SELECT AVG(salary) FROM doctor

Q62. Total fines per member:

```
SELECT m.name, [Box1] AS total FROM member m JOIN loan l ... JOIN fine f ...
[Box2] m.memberID;
```

Answer

Box1: SUM(f.amount) — Box2: GROUP BY

Q63. Update Electronics price +10%:

```
[Box1] product [Box2] price = price * 1.10 WHERE category = 'Electronics';
```

Answer

Box1: UPDATE — Box2: SET

Q64. Delete returned loans:

```
[Box1] FROM loan WHERE returnDate [Box2];
```

Answer

Box1: DELETE — Box2: IS NOT NULL

Q65. Create view of unpaid fines:

```
[Box1] VIEW unpaid_fines [Box2] SELECT ...;
```

Answer

Box1: CREATE — Box2: AS

Q66. Insert book:

```
[Box1] INTO book [Box2] (101, 'DB Systems', 'Date', 'Tech', 2020);
```

Answer

Box1: INSERT — Box2: VALUES

Q67. Members with avg fine > overall avg:

```
GROUP BY m.memberID [Box1] AVG(f.amount) > ([Box2]);
```

Answer

Box1: HAVING — Box2: SELECT AVG(amount) FROM fine

Q68. Products never ordered:

```
SELECT p.name FROM product p [Box1] JOIN orderItem oi ON ... WHERE oi.prodID [Box2];
```

Answer

Box1: LEFT — Box2: IS NULL

Q69. Highest paid per specialty:

```
WHERE salary = (SELECT [Box1] FROM doctor d2 WHERE [Box2]);
```

Answer

Box1: MAX(salary) — Box2: d1.specialty = d2.specialty

Q70. Customers ordering same product multiple times:

```
GROUP BY ... [Box1] COUNT(*) > 1;
```

Answer

Box1: HAVING

Q71. Revenue per category:

```
SELECT category, [Box1] AS revenue ... GROUP BY [Box2];
```

Answer

Box1: SUM(quantity * unitPrice) — Box2: category

Q72. Add FK constraint:

```
ALTER TABLE loan ADD [Box1] (memberID) [Box2] member(memberID);
```

Answer

Box1: FOREIGN KEY — Box2: REFERENCES

Q73. Genres with all books after 2010:

```
HAVING [Box1](publishYear) [Box2] 2010;
```

Answer

Box1: MIN — Box2: >

Q74. Doctors never had appointment:

```
WHERE [Box1] (SELECT 1 FROM appointment a WHERE [Box2]);
```

Answer

Box1: NOT EXISTS — Box2: a.docID = d.docID

Q75. Members with fines in both 2023 AND 2024:

```
... WHERE YEAR(loanDate) = 2023 [Box] ... WHERE YEAR(loanDate) = 2024;
```

Answer

Box: INTERSECT

Q76. Books borrowed more than 5 times:

```
SELECT title, [Box1] as cnt ... [Box2] b.bookID HAVING COUNT(*) > 5;
```

Answer

Box1: COUNT(*) — Box2: GROUP BY

Q77. Average order per country:

```
SELECT country, [Box1] ... GROUP BY [Box2];
```

Answer

Box1: AVG(totalAmount) — Box2: country

Q78. Doctors treating all blood types:

```
WHERE (SELECT COUNT([Box1] bloodType) ... WHERE a.docID=d.docID) = (SELECT COUNT([
    Box2] bloodType) FROM patient);
```

Answer

Box1: DISTINCT — Box2: DISTINCT

Q79. Delete appointments before 2020:

```
[Box1] FROM appointment WHERE appointDate [Box2] '2020-01-01';
```

Answer

Box1: DELETE — Box2: <

Q80. Top 3 borrowed books:

```
... GROUP BY b.bookID [Box1] cnt [Box2] LIMIT 3;
```

Answer

Box1: ORDER BY — Box2: DESC

Q81. Patients above gender average age:

```
WHERE age > (SELECT [Box1] FROM patient p2 WHERE p2.gender [Box2] p1.gender);
```

Answer

Box1: AVG(age) — Box2: =

Q82. Create index:

```
[Box1] INDEX idx_date [Box2] appointment(appointDate);
```

Answer

Box1: CREATE — Box2: ON

Q83. Members who borrowed all J.K. Rowling books:

```
WHERE NOT EXISTS (SELECT bookID FROM book WHERE author='J.K. Rowling' AND [Box1] (
    SELECT * FROM loan WHERE memberID=m.memberID AND bookID [Box2] b.bookID));
```

Answer

Box1: NOT EXISTS — Box2: =

Q84. Products ordered from all countries:

```
WHERE (SELECT COUNT(DISTINCT country) ... WHERE prodID [Box1] p.prodID) = (SELECT
    COUNT([Box2] country) FROM customer);
```

Answer

Box1: = — Box2: DISTINCT

Q85. Customers NOT ordering products over 500:

```
WHERE custID [Box1] (SELECT custID ... WHERE price [Box2] 500);
```

Answer

Box1: NOT IN — Box2: >

Q86. Restaurants with Italian AND Chinese:

```
SELECT rid ... WHERE type='Italian' [Box] SELECT rid ... WHERE type='Chinese';
```

Answer

Box: INTERSECT

Q87. COALESCE for NULL tip:

```
SELECT [Box](tip, 0) FROM orders;
```

Answer

Box: COALESCE

Q88. CTE syntax:

```
[Box1] high_salary AS (SELECT ...) [Box2] * FROM high_salary;
```

Answer

Box1: WITH — Box2: SELECT

Q89. Count patients per blood type with >10:

```
GROUP BY bloodType [Box] COUNT(*) > 10;
```

Answer

Box: HAVING

Q90. Find NULL emails:

```
WHERE email [Box];
```

Answer

Box: IS NULL

Q91. Names NOT ending with .com:

```
WHERE email [Box] LIKE '\%.com';
```

Answer

Box: NOT

Q92. All books + loans (including never borrowed):

```
SELECT * FROM book b [Box] JOIN loan l ON b.bookID = l.bookID;
```

Answer

Box: LEFT

Q93. CASE for age categories:

```
SELECT name, [Box] WHEN age<18 THEN 'Minor' WHEN age<65 THEN 'Adult' ELSE 'Senior'
    END FROM patient;
```

Answer

Box: CASE

Q94. Count distinct categories ordered:

```
SELECT COUNT([Box] category) FROM product p JOIN orderItem oi ON...;
```

Answer

Box: DISTINCT

Q95. MIN and MAX age by department:

```
SELECT dept, [Box1], [Box2] FROM employee GROUP BY dept;
```

Answer

Box1: MIN(age) — Box2: MAX(age)

Q96. Sum salaries where salary > avg:

```
SELECT SUM(salary) FROM employee WHERE salary > ([Box]);
```

Answer

Box: SELECT AVG(salary) FROM employee

Q97. Full outer join both tables:

```
SELECT * FROM A [Box] OUTER JOIN B ON A.id = B.id;
```

Answer

Box: FULL

Q98. Rename column in output:

```
SELECT name [Box] emp_name FROM employee;
```

Answer

Box: AS

Q99. Count all rows including NULL:

```
SELECT [Box] FROM patient;
```

Answer

Box: COUNT(*)

Q100. Check if subquery returns at least one row:

```
WHERE [Box] (SELECT * FROM appointment WHERE...);
```

Answer

Box: EXISTS

Quick Reference

SQL Execution Order: FROM → WHERE → GROUP BY → HAVING → SELECT → ORDER BY

Join Types: INNER (only matches), LEFT (all left), RIGHT (all right), FULL (all both), CROSS (Cartesian), NATURAL (auto common cols)

RA Patterns:

- Find X where Y: $\Pi_X(\sigma_Y(R))$
- NOT/never: $\Pi_{attr}(R) - \Pi_{attr}(\sigma_{cond}(R))$
- ALL (division): $R \div S$
- OR: $\cup$ AND: $\cap$

Common Mistakes:

1. = NULL instead of IS NULL
2. Aggregates in WHERE (use HAVING)
3. Non-grouped cols in SELECT without GROUP BY
4. Missing join condition (Cartesian product)
5. Double quotes for strings (use single)
6. $R - S \neq S - R$